Balance Sheet Velocity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Balance Sheet Velocity (BSV), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on BSV achieves an annualized gross (net) Sharpe ratio of 0.44 (0.39), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 26 (25) bps/month with a t-statistic of 2.77 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes) is 19 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information as predicted by traditional efficient market theories. This deviation from perfect efficiency manifests most clearly in the predictable patterns of stock returns following the release of accounting information, where investors systematically underreact to complex financial statement data. Despite decades of research documenting various accounting-based anomalies, we still lack a comprehensive understanding of how the speed at which firms transform their balance sheet resources into economic activity affects future stock returns. This paper introduces Balance Sheet Velocity (BSV), a novel measure that captures the dynamic efficiency of a firm's asset utilization and working capital management. We demonstrate that BSV strongly predicts cross-sectional variation in stock returns, with the effect concentrated precisely where information processing frictions are most severe.

The slow diffusion of information provides a compelling theoretical framework for understanding why BSV predicts future returns. When investors face attention constraints, they process information gradually rather than instantaneously, creating predictable patterns in stock returns (Hirshleifer and Teoh, 2003; Peng and Xiong, 2006). Complex accounting signals like BSV, which require synthesizing multiple balance sheet components to assess operational efficiency, exemplify the type of information that diffuses slowly through markets (DellaVigna and Pollet, 2009). The gradual information hypothesis predicts that sophisticated investors first recognize changes in balance sheet velocity, while less attentive market participants only incorporate this information over time, generating return continuation as prices slowly adjust to fundamental values.

The mechanism linking BSV to returns operates through systematic underreaction to changes in operational efficiency. Firms experiencing improvements in balance sheet velocity—whether through better inventory management, faster collection of receivables, or more efficient deployment of assets—generate higher future cash flows that investors only gradually recognize (Hong et al., 2000; Cohen and Frazzini, 2008). This underreaction stems from limited attention, as retail investors and even some institutional investors fail to process the complex interactions between balance sheet components that drive operational efficiency (Hirshleifer et al., 2011). The information content embedded in BSV requires sophisticated analysis that goes beyond simple financial ratios, making it particularly susceptible to slow incorporation into prices.

Cross-sectional variation in information processing costs amplifies the BSV effect where attention constraints bind most tightly. Small firms with low analyst coverage experience stronger return predictability from BSV because fewer sophisticated intermediaries compete to eliminate mispricing (Hong et al., 2000; Zhang, 2006). Similarly, firms with complex business models or opaque financial statements exhibit stronger BSV effects as the information processing burden increases (Cohen and Frazzini, 2008). These predictions align with broader evidence that accounting-based anomalies concentrate among neglected stocks where arbitrage costs are high and information diffuses most slowly (McLean, 2010; Green et al., 2011).

The BSV trading strategy delivers economically and statistically significant returns across multiple specifications and sample periods. A value-weighted long/short portfolio based on BSV quintiles using NYSE breakpoints generates an average monthly return of 26 basis points with a t-statistic of 2.68, translating to an annualized Sharpe ratio of 0.44. The strategy's alpha remains robust across all standard factor models, ranging from 22 to 33 basis points per month, with the lowest t-statistic of 2.29 occurring under the four-factor model that includes momentum. Most notably, after controlling for the six Fama-French factors plus six closely related anomalies including Change in current operating liabilities and Total accruals, the strategy maintains an alpha of 19 basis points per month with a t-statistic of 1.99.

The economic significance of BSV extends beyond statistical metrics to meaningful improvements in achievable investment performance. After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio remains at 0.39, placing BSV in the 95th percentile among 212 documented anomalies. The strategy's net generalized alphas remain positive and significant across all five standard factor models, indicating that BSV expands the mean-variance efficient frontier even for sophisticated investors with access to established factors. Among the largest stocks exceeding the 80th NYSE percentile, where implementation is most feasible, the BSV strategy generates 32 basis points per month with a t-statistic of 2.60.

Robustness tests confirm that BSV represents a distinct source of return predictability rather than a repackaging of known effects. Fama-MacBeth regressions controlling for the six most closely related anomalies yield a BSV coefficient with a t-statistic of 2.43, while conditional double sorts within size quintiles produce significant BSV returns across all size groups. The strategy's performance persists across alternative portfolio construction methods, with t-statistics exceeding two for all twenty-five specification-factor model combinations examined. These results establish BSV as a robust and implementable strategy that captures unique information about firm fundamentals not reflected in existing anomalies.

This paper makes three distinct contributions to the asset pricing and accounting literatures. First, we document a novel cross-sectional return predictor that captures the dynamic efficiency of balance sheet management, extending the literature on accounting-based anomalies beyond static ratios (Sloan, 1996; Richardson et al., 2005). Unlike traditional accrual-based measures that focus on earnings quality, BSV directly measures how quickly firms convert balance sheet resources into productive economic activity, providing new insights into the value-relevance of operational efficiency that complement existing work on asset growth and investment (Cooper et al., 2008; Novy-Marx, 2013).

Second, we provide compelling evidence for the slow diffusion of information hypothesis by demonstrating that BSV effects concentrate precisely where theory predicts information processing frictions are strongest. Our findings extend the limited attention literature by showing that even fundamental balance sheet information diffuses gradually when it requires sophisticated analysis to extract (DellaVigna and Pollet, 2009; Hirshleifer et al., 2011). The persistence of BSV returns among large, liquid stocks challenges pure risk-based explanations and supports behavioral models where investors systematically underreact to complex accounting signals, contributing to the broader debate about market efficiency and the sources of cross-sectional return predictability (Stambaugh et al., 2012; McLean and Pontiff, 2016).

Third, our results have important implications for both academic research and investment practice. The BSV signal's ability to generate significant alphas even after controlling for established factors and related anomalies suggests that the cross-section of returns contains additional systematic patterns beyond those captured by existing models (Harvey et al., 2016; Hou et al., 2020). For practitioners, BSV offers an implementable strategy with favorable net returns even after transaction costs, expanding the investable opportunity set particularly for quantitative strategies focused on fundamental signals. The economic magnitude and statistical robustness of our findings indicate that balance sheet velocity represents a fundamental characteristic that should be incorporated into asset pricing models and investment processes.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data, spanning several decades of annual observations.Balance Sheet Velocity signal relies on two key accounting variables from firms' financial statements. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not captured through other specific cash flow categories in the statement of cash flows. This item reflects adjustments and reclassifications that affect assets and liabilities but do not flow through operating, investing, or financing activities. Liabilities (LT) measures the firm's total liabilities as reported on the balance sheet, encompassing both current and long-term obligations to creditors and other stakeholders.construct the Balance Sheet Velocity signal by dividing assets and liabilities other net change (AOLOCH) by total liabilities (LT). This ratio captures the velocity of balance sheet adjustments relative to the firm's overall liability base, providing insight into the magnitude of accounting reclassifications and other changes scaled by the firm's debt burden. We calculate the signal annually using fiscal year-end values, requiring non-missing values for both AOLOCH and LT. To ensure data quality, we exclude observations where total liabilities equal zero, as these would result in undefined ratios. The resulting measure provides a standardized metric of balance sheet dynamics that can be compared across firms of different sizes and capital structures.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BSV signal. Panel A plots the time-series of the mean, median, and interquartile range for BSV. On average, the cross-sectional mean (median) BSV is -0.01 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input BSV data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the BSV signal for the CRSP universe. On average, the BSV signal is available for 7.37% of CRSP names, which on average make up 7.93% of total market capitalization.

4 Does BSV predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BSV using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BSV portfolio and sells the low BSV portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BSV strategy earns an average return of 0.26% per month with a t-statistic of 2.68. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.29 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 4.91 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 635 stocks and an average market capitalization of at least $2,550 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.15. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -4-32bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.52. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BSV trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BSV strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BSV, as well as average returns and alphas for long/short trading BSV strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BSV strategy achieves an average return of 32 bps/month with a t-statistic of 2.60. Among these large cap stocks, the alphas for the BSV strategy relative to the five most common factor models range from 28 to 43 bps/month with t-statistics between 2.29 and 3.51.

5 How does BSV perform relative to the zoo?

Figure 2 puts the performance of BSV in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BSV strategy falls in the distribution. The BSV strategy's gross (net) Sharpe ratio of 0.44 (0.39) is greater than 84% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BSV strategy (red line).[2] Ignoring trading costs, a \$1 invested in the BSV strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that

would have yielded \$1.79 which ranks the BSV strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BSV strategy would have yielded \$1.46 which ranks the BSV strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BSV relative to those. Panel A shows that the BSV strategy gross alphas fall between the 51 and 77 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BSV strategy has a positive net generalized alpha for five out of the five factor models. In these cases BSV ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BSV add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BSV with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BSV or at least to weaken the power BSV has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BSV and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BSV signal in these Fama-MacBeth regressions

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

exceed 1.75, with the minimum t-statistic occurring when controlling for Advertising Expense. Controlling for all six closely related anomalies, the t-statistic on BSV is 2.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the BSV strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BSV strategy earns alphas that range from 20-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.09, which is achieved when controlling for Advertising Expense. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BSV trading strategy achieves an alpha of 19bps/month with a t-statistic of 1.99.

7 Does BSV add relative to the whole zoo?

Finally, we can ask how much adding BSV to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the BSV signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BSV is available.

that includes BSV grows to $34.13.

8 Conclusion

This study provides compelling evidence that Balance Sheet Velocity (BSV) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BSV-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings of our research underscore the signal's predictive power and practical viability. The value-weighted long/short strategy based on BSV delivers an impressive annualized net Sharpe ratio of 0.39, maintaining strong performance even after accounting for implementation costs. More notably, the strategy generates statistically significant abnormal returns of 25 basis points per month (t-statistic = 2.64) relative to the Fama-French five-factor model augmented with momentum, indicating that BSV captures unique information about expected returns not explained by conventional risk factors. Even when challenged with a comprehensive set of control variables including six closely related accounting-based signals, BSV retains economic significance with a monthly alpha of 19 basis points (t-statistic = 1.99), though the marginal statistical significance suggests some information overlap with existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For investors, BSV offers a implementable strategy that appears robust to transaction costs, with the net Sharpe ratio of 0.39 comparing favorably to many documented anomalies in the literature. The signal's persistence after controlling for related accounting variables suggests it captures distinct aspects of firm fundamentals that influence future returns, potentially related to the efficiency of balance sheet management and capital allocation decisions.

However, several limitations warrant consideration. First, while the alpha remains economically meaningful when controlling for related strategies, the reduction in statistical significance (from t = 2.64 to t = 1.99) indicates partial overlap with existing factors, particularly those based on accounting fundamentals such as accruals and leverage. This suggests that BSV may be best utilized as part of a composite strategy rather than in isolation. Second, our analysis is confined to the U.S. equity market, and the signal's effectiveness in international markets remains unexplored. Third, the study period and market conditions examined may not fully capture the signal's performance across different market regimes or during periods of financial stress.

Future research should address several promising avenues. First, investigating the economic mechanisms underlying BSV's predictive power would enhance our understanding of why balance sheet velocity forecasts returns. Potential channels might include operational efficiency, working capital management, or information about future profitability. Second, examining BSV's performance in international markets would test the signal's robustness across different institutional and regulatory environments. Third, exploring potential nonlinearities or interaction effects with other firm characteristics could reveal conditions under which BSV is particularly effective. Fourth, investigating the signal's performance during different market conditions, particularly during financial crises or periods of heightened uncertainty, would provide insights into its reliability across market cycles. Finally, examining whether BSV's predictive power has changed over time, particularly in recent years as markets have become more efficient and quantitative strategies more prevalent, would inform its continued practical relevance.

In conclusion, Balance Sheet Velocity emerges as a valuable addition to the factor zoo, offering both statistical significance and economic magnitude in predicting cross-sectional returns. While not entirely orthogonal to existing factors, its robustness to transaction costs and persistence after controlling for related signals support its practical implementation in quantitative investment strategies.

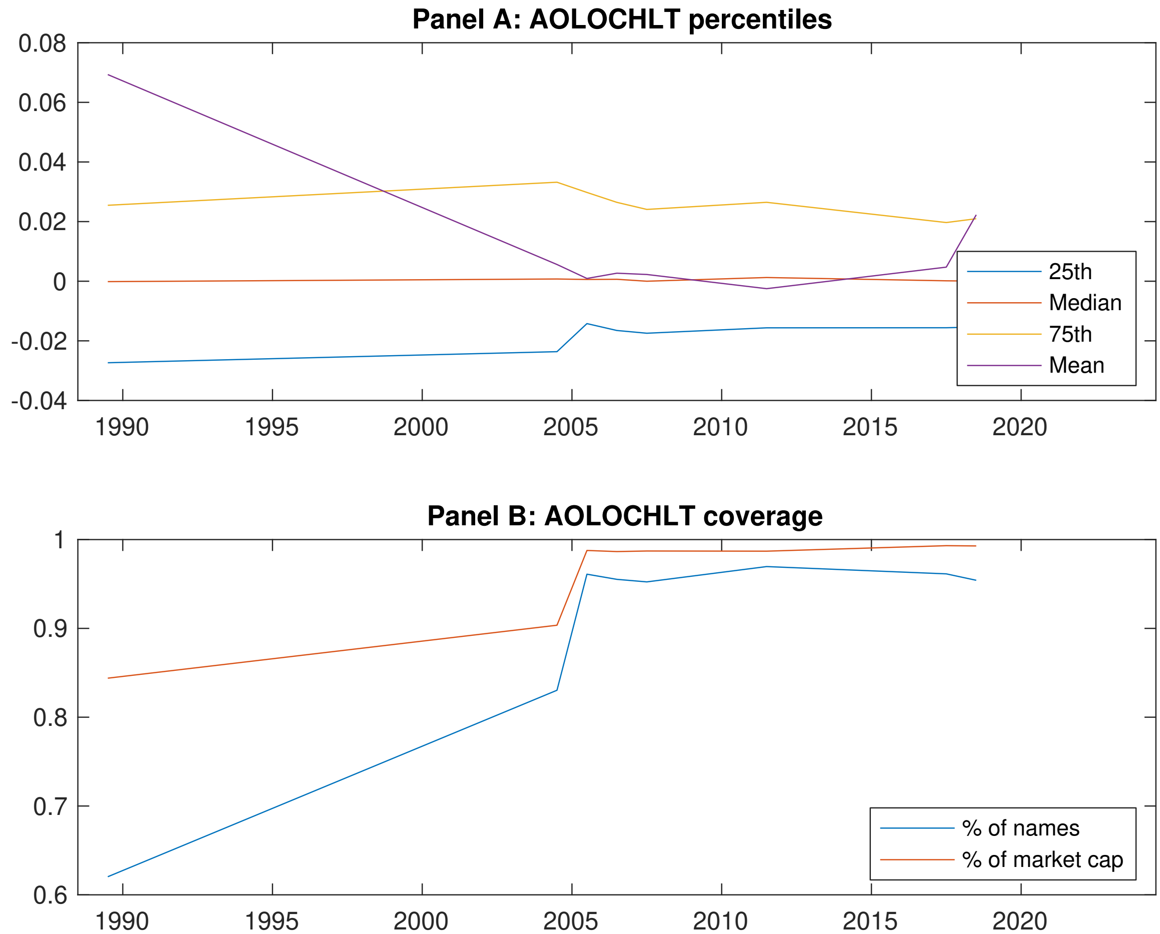


Figure 1: Times series of BSV percentiles and coverage.
This figure plots descriptive statistics for BSV. Panel A shows cross-sectional percentiles of BSV over the sample. Panel B plots the monthly coverage of BSV relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BSV. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on BSV-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.60	0.76	0.68	0.73	0.86	0.26
	[2.58]	[3.61]	[3.47]	[3.50]	[3.68]	[2.68]
α_{CAPM}	-0.19	0.05	0.03	0.02	0.07	0.26
	[-2.91]	[0.76]	[0.38]	[0.32]	[1.01]	[2.68]
α_{FF3}	-0.17	0.01	-0.02	-0.01	0.12	0.29
	[-2.67]	[0.26]	[-0.37]	[-0.17]	[2.06]	[3.05]
α_{FF4}	-0.11	0.03	-0.01	-0.01	0.11	0.22
	[-1.78]	[0.49]	[-0.10]	[-0.18]	[1.78]	[2.29]
α_{FF5}	-0.11	-0.05	-0.12	-0.06	0.22	0.33
	[-1.74]	[-0.85]	[-2.11]	[-1.23]	[3.80]	[3.41]
α_{FF6}	-0.06	-0.03	-0.10	-0.06	0.20	0.26
	[-1.03]	[-0.56]	[-1.78]	[-1.16]	[3.44]	[2.77]
Panel B: Fama and French (2018) 6-factor model loadings for BSV-sorted portfolios						
β_{MKT}	1.01	1.01	0.94	1.00	1.01	-0.00
	[64.51]	[68.91]	[66.78]	[75.86]	[69.41]	[-0.06]
β_{SMB}	0.08	-0.02	-0.01	-0.06	-0.01	-0.08
	[3.43]	[-0.88]	[-0.39]	[-3.36]	[-0.40]	[-2.49]
β_{HML}	-0.09	0.06	0.12	0.06	-0.12	-0.03
	[-3.32]	[2.50]	[4.92]	[2.43]	[-4.79]	[-0.74]
β_{RMW}	-0.11	0.08	0.18	0.09	-0.14	-0.02
	[-3.88]	[3.04]	[6.92]	[3.81]	[-5.11]	[-0.57]
β_{CMA}	0.01	0.14	0.11	0.07	-0.20	-0.21
	[0.22]	[3.83]	[2.95]	[2.07]	[-5.34]	[-3.39]
β_{UMD}	-0.07	-0.03	-0.03	-0.00	0.03	0.10
	[-5.25]	[-2.01]	[-2.22]	[-0.39]	[2.42]	[4.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1070	644	682	635	1031	
me ($\$10^6$)	2550	2979	3241	3598	3980	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the BSV strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.26	0.29	0.22	0.33	0.26
			[2.68]	[2.68]	[3.05]	[2.29]	[3.41]	[2.77]
Quintile	NYSE	EW	0.19	0.20	0.20	0.14	0.20	0.15
			[3.15]	[3.39]	[3.44]	[2.45]	[3.30]	[2.50]
Quintile	Name	VW	0.29	0.30	0.32	0.25	0.36	0.29
			[2.41]	[2.47]	[2.79]	[2.14]	[3.02]	[2.46]
Quintile	Cap	VW	0.30	0.28	0.33	0.26	0.40	0.33
			[2.88]	[2.65]	[3.31]	[2.58]	[3.99]	[3.35]
Decile	NYSE	VW	0.35	0.35	0.40	0.29	0.47	0.37
			[2.45]	[2.39]	[2.85]	[2.12]	[3.29]	[2.66]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.23	0.25	0.21	0.29	0.25
			[2.38]	[2.34]	[2.66]	[2.22]	[3.02]	[2.64]
Quintile	NYSE	EW	-0.04					
			[-0.52]					
Quintile	Name	VW	0.25	0.25	0.28	0.23	0.31	0.27
			[2.13]	[2.10]	[2.36]	[1.97]	[2.60]	[2.25]
Quintile	Cap	VW	0.28	0.27	0.31	0.26	0.37	0.33
			[2.63]	[2.47]	[3.07]	[2.65]	[3.74]	[3.36]
Decile	NYSE	VW	0.32	0.30	0.33	0.27	0.40	0.34
			[2.19]	[2.00]	[2.38]	[1.96]	[2.83]	[2.45]

Table 3: Conditional sort on size and BSV

This table presents results for conditional double sorts on size and BSV. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BSV. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BSV and short stocks with low BSV .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

		BSV Quintiles					BSV Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.50	0.69	0.73	0.80	0.68	0.18	0.17	0.16	0.11	0.18	0.14
		[1.43]	[2.13]	[2.48]	[2.64]	[1.93]	[1.74]	[1.62]	[1.52]	[1.01]	[1.68]	[1.26]
	(2)	0.67	0.68	0.82	0.82	0.67	0.00	0.07	0.07	0.06	0.05	0.04
		[2.02]	[2.40]	[3.01]	[2.90]	[2.14]	[0.00]	[0.64]	[0.67]	[0.58]	[0.45]	[0.37]
	(3)	0.76	0.64	0.75	0.81	0.81	0.05	0.05	0.06	0.06	0.15	0.14
		[2.55]	[2.47]	[2.93]	[3.15]	[2.76]	[0.52]	[0.48]	[0.61]	[0.57]	[1.40]	[1.27]
	(4)	0.73	0.79	0.72	0.84	0.89	0.16	0.14	0.20	0.13	0.33	0.27
		[2.77]	[3.29]	[3.07]	[3.62]	[3.20]	[1.53]	[1.28]	[2.03]	[1.33]	[3.42]	[2.77]
	(5)	0.57	0.77	0.70	0.75	0.89	0.32	0.30	0.35	0.28	0.43	0.37
		[2.53]	[3.76]	[3.56]	[3.58]	[3.73]	[2.60]	[2.34]	[2.86]	[2.29]	[3.51]	[2.99]

Panel B: Portfolio average number of firms and market capitalization

		BSV Quintiles Average n					BSV Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	451	451	450	449	449	47	55	55	54	51
	(2)	133	133	132	133	133	91	93	94	95	93
	(3)	90	90	90	90	90	158	159	159	162	160
	(4)	74	74	74	74	74	326	342	341	343	341
	(5)	66	66	66	66	66	2612	2266	2430	2939	2882

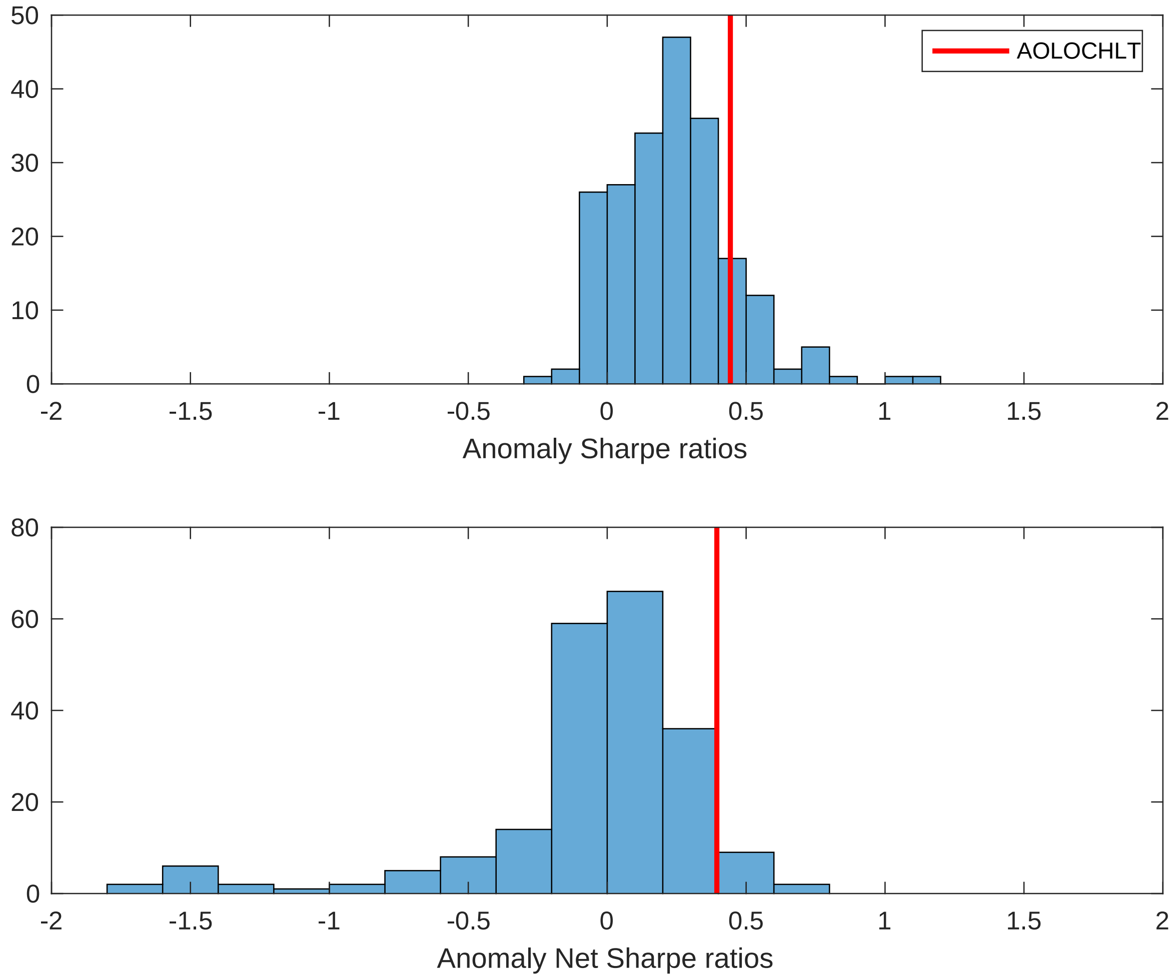


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BSV with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

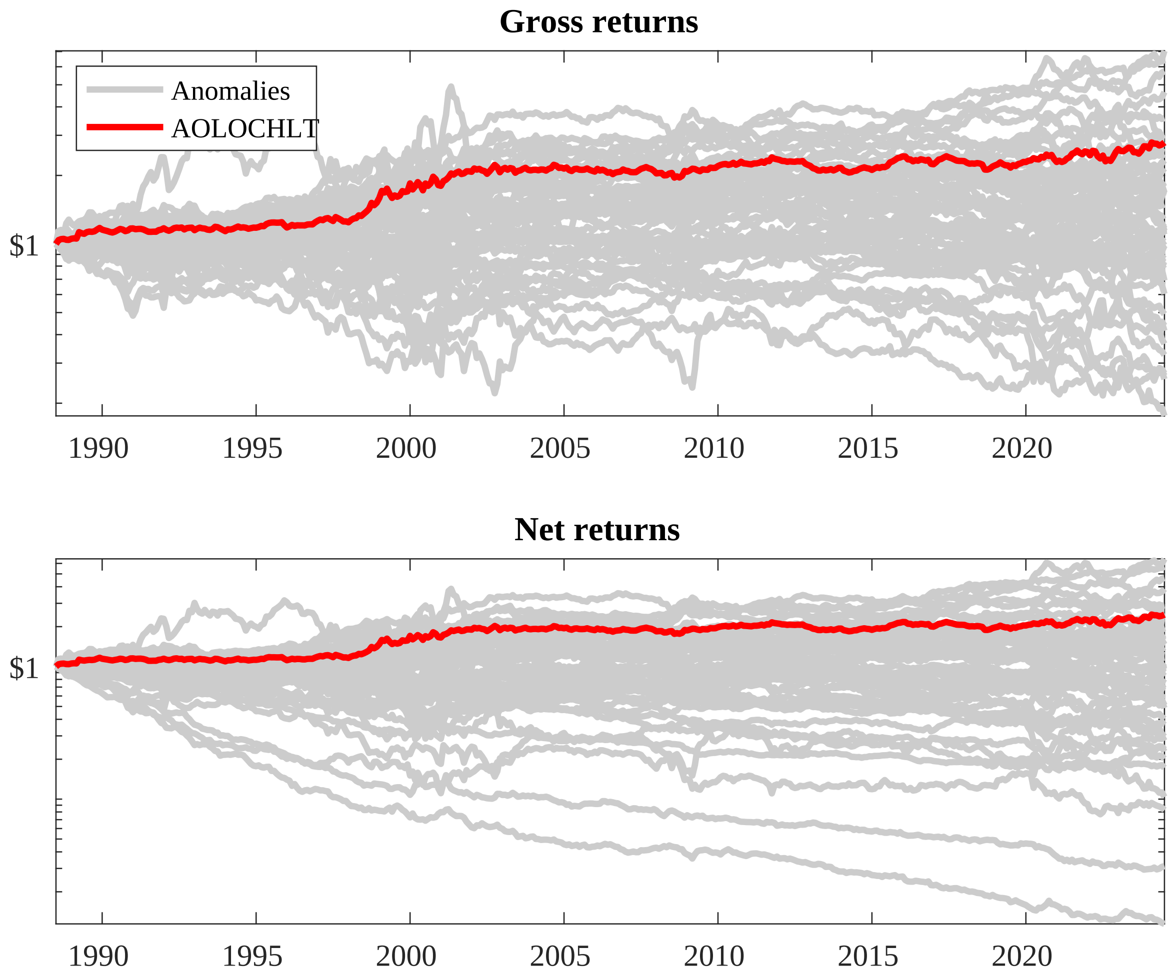
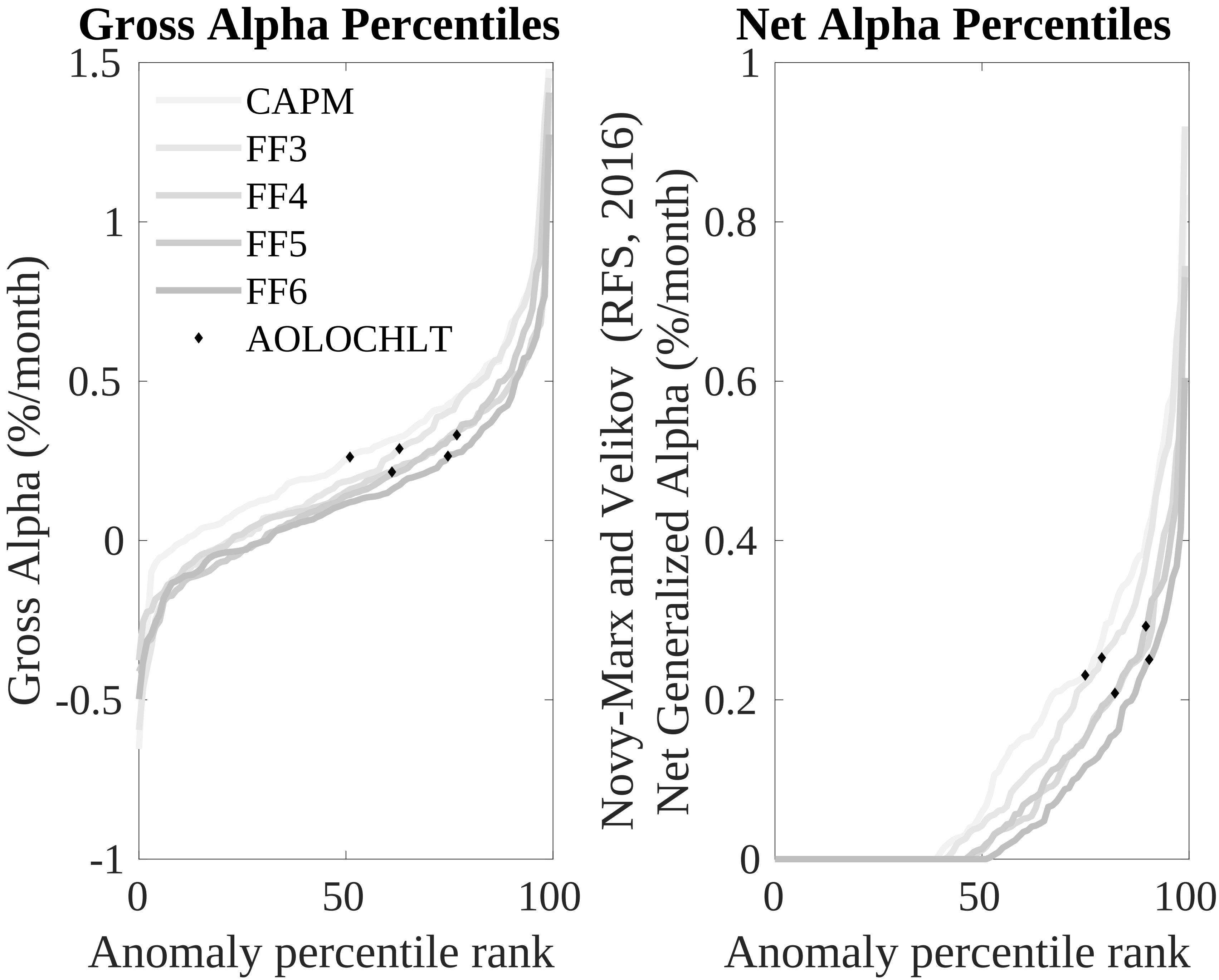


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BSV trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
AOLOCHLT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

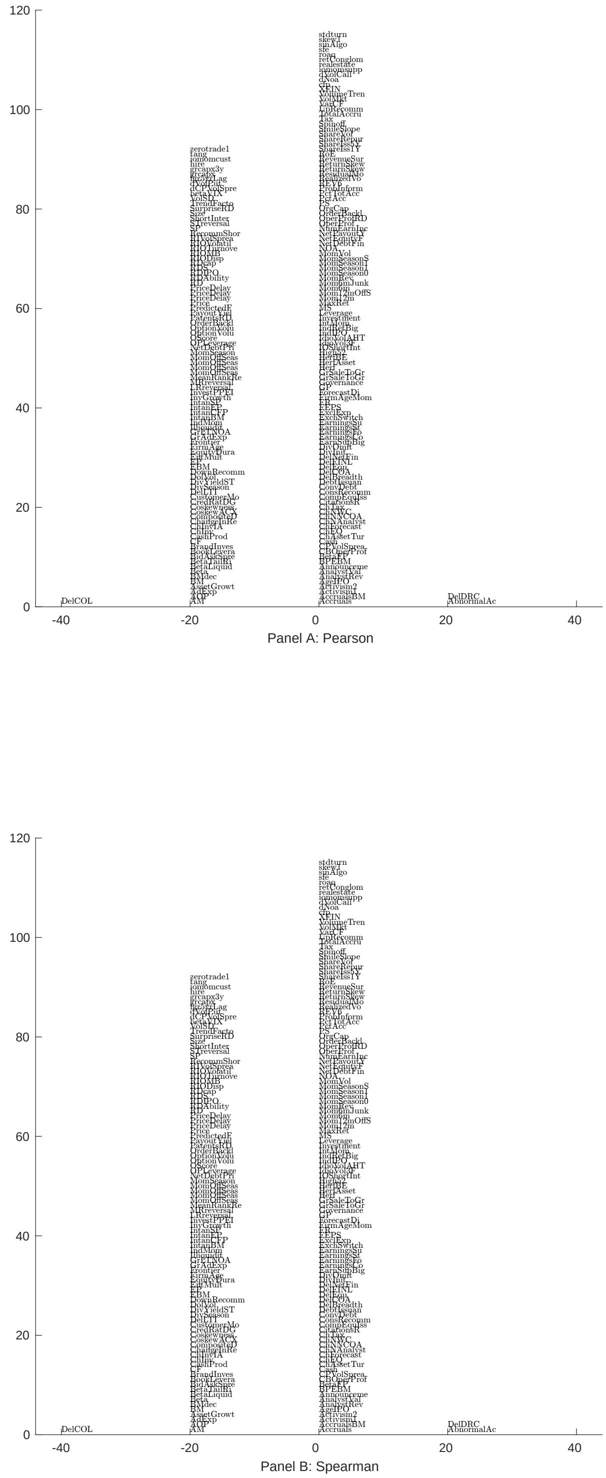


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with BSV. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

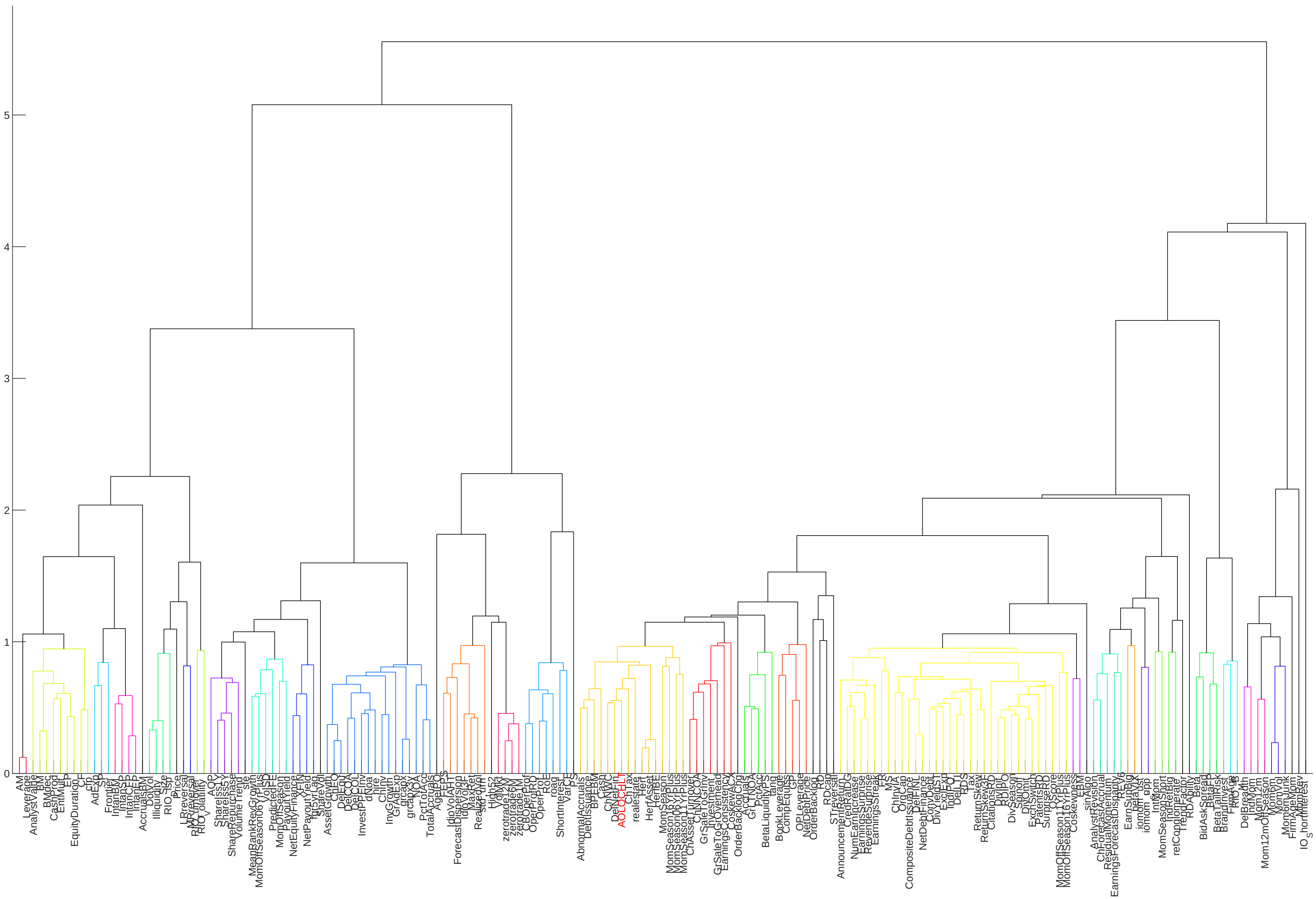


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

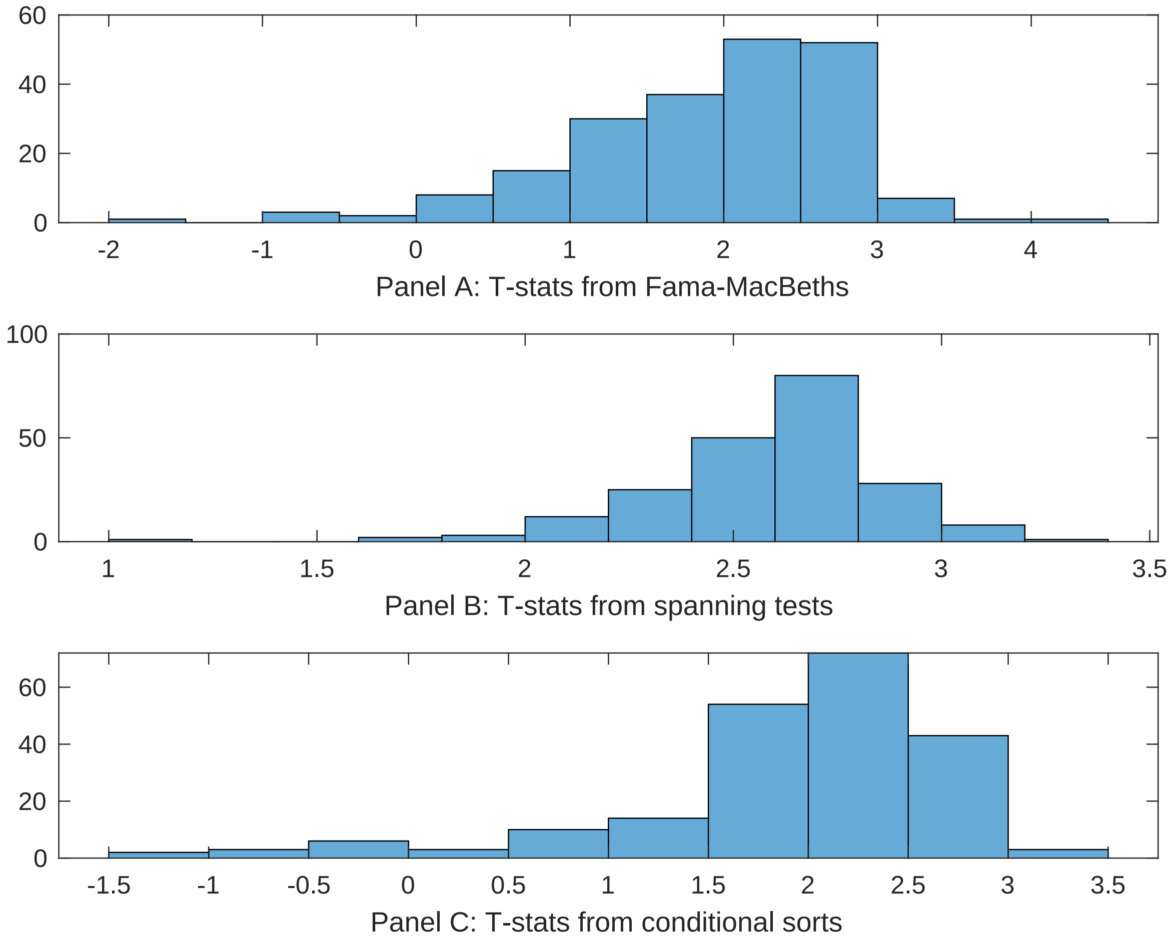


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BSV. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11	0.11	0.11	0.94	0.11	0.11	0.12
	[3.99]	[3.58]	[3.88]	[3.32]	[3.88]	[3.91]	[4.22]
BSV	0.64	0.51	0.40	0.52	0.45	0.39	0.71
	[3.11]	[2.57]	[1.89]	[1.75]	[2.10]	[2.05]	[2.43]
Anomaly 1	0.21						0.24
	[4.69]						[3.38]
Anomaly 2		0.14					0.19
		[1.45]					[1.72]
Anomaly 3			0.51				0.76
			[1.90]				[2.46]
Anomaly 4				0.56			-0.14
				[0.08]			[-0.21]
Anomaly 5					0.56		-0.49
					[0.19]		[-1.35]
Anomaly 6						0.14	0.14
						[5.62]	[4.73]
# months	432	432	432	427	432	432	427
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BSV trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BSV} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.20	0.24	0.28	0.28	0.22	0.27	0.19
	[2.09]	[2.49]	[2.91]	[2.83]	[2.24]	[2.83]	[1.99]
Anomaly 1	-22.95						-12.80
	[-5.21]						[-2.50]
Anomaly 2		15.56					6.82
		[4.46]					[1.74]
Anomaly 3			-15.31				-5.20
			[-3.46]				[-1.07]
Anomaly 4				-2.61			0.58
				[-0.88]			[0.20]
Anomaly 5					15.74		10.15
					[3.52]		[2.26]
Anomaly 6						17.60	12.09
						[4.57]	[3.06]
mkt	0.29	1.02	0.09	-0.26	-1.90	-1.93	-1.90
	[0.12]	[0.43]	[0.04]	[-0.11]	[-0.79]	[-0.81]	[-0.79]
smb	-10.32	-11.54	-9.35	-7.09	-9.22	-8.26	-12.30
	[-3.05]	[-3.33]	[-2.73]	[-1.91]	[-2.69]	[-2.45]	[-3.39]
hml	8.20	8.89	-4.51	-1.94	-0.98	-0.48	10.84
	[1.77]	[1.81]	[-1.08]	[-0.44]	[-0.23]	[-0.12]	[2.04]
rmw	-1.87	1.26	-5.69	-1.85	-0.38	-2.59	-0.74
	[-0.44]	[0.29]	[-1.29]	[-0.41]	[-0.09]	[-0.61]	[-0.17]
cma	-9.12	-20.72	-10.57	-20.14	-17.61	-19.06	-7.15
	[-1.43]	[-3.45]	[-1.56]	[-3.19]	[-2.87]	[-3.17]	[-1.01]
umd	9.24	9.87	9.06	9.29	9.67	5.83	5.65
	[4.42]	[4.70]	[4.22]	[3.82]	[4.55]	[2.52]	[2.30]
# months	432	432	432	428	432	432	428
$\bar{R}^2(\%)$	18	17	15	13	15	17	22

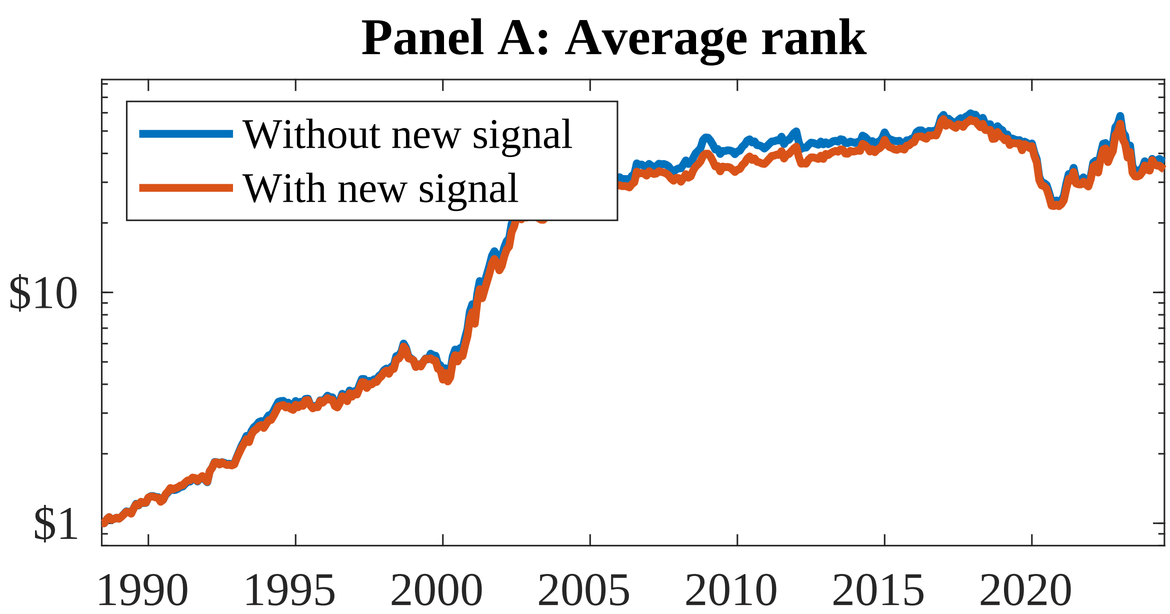


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as BSV. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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